FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Acute Lower Respiratory Tract Infections

Tutorial | Week 1 · Breathing, fever & the clinical methods | Paediatrics | Dr Nabatte

Pneumonia is the biggest infectious killer of Ugandan children; it is diagnosed by counting breaths, looking for chest indrawing and danger signs, and measuring $SpO_2$, then classified by WHO/IMCI into pneumonia (oral amoxicillin at home) or severe pneumonia (oxygen, ampicillin or penicillin plus gentamicin in hospital), while bronchiolitis, pertussis, empyema, aspiration and HIV-related pneumonia each need their own recognition and treatment.

**Must know before the ward round**

1. Fast breathing (WHO/IMCI): ≥60/min under 2 months; ≥50/min at 2–11 months; ≥40/min at 1–4 years; UCG 2023: >30/min from 5 years. Count for a full 60 seconds in a calm child.
2. Pneumonia (WHO 2014/UCG 2023/WHO 2024) = cough or difficult breathing + fast breathing and/or chest indrawing, no danger sign → oral amoxicillin 40 mg/kg twice daily (80 mg/kg/day) for 5 days at home (3 days only for fast-breathing pneumonia in low-HIV areas; Uganda is high-HIV, so 5 days).
3. Severe pneumonia = cough/difficult breathing + any general danger sign (unable to drink, vomits everything, convulsions, lethargic/unconscious), central cyanosis or $SpO_2$ <90%, or severe respiratory distress (grunting, very severe indrawing) → admit, oxygen, ampicillin 50 mg/kg (or benzylpenicillin 50,000 IU/kg) IV/IM 6-hourly + gentamicin 7.5 mg/kg once daily for at least 5 days.
4. Every infant under 2 months with pneumonia is treated as severe (UCG 2023): ampicillin + gentamicin (5 mg/kg in the first week of life).
5. No better after 48 h on first line → ceftriaxone 80 mg/kg once daily; suspect Staph (pneumatocoeles, empyema) → cloxacillin 50 mg/kg 6-hourly + gentamicin for 3 weeks; think TB, HIV/PCP, empyema, foreign body, heart failure.
6. Oxygen for $SpO_2$ <90% (WHO); target ≥94% if emergency signs. Nasal prongs 0.5–1 L/min neonates, 1–2 L/min infants, 2–4 L/min older children. Without an oximeter: central cyanosis, grunting, head nodding, nasal flaring, inability to drink, RR ≥70 suggest hypoxaemia.
7. Bronchiolitis: under 2 years (peak 2–6 months), RSV, coryza then cough, fast breathing, crackles and wheeze. Supportive care only: oxygen if $SpO_2$ <90–92%, small frequent or NG feeds, gentle nasal suction. No routine antibiotics, steroids or nebulised therapy (NICE NG9); UCG 2023 allows a salbutamol trial, stopped if no response.
8. HIV-exposed/infected infant with severe pneumonia: add high-dose cotrimoxazole for PCP (UCG: 120 mg/kg/day in 3–4 doses for 21 days) + prednisolone if hypoxic; start or continue ART.
9. Pertussis: catarrhal → paroxysmal (whoop, post-tussive vomiting, apnoea and cyanosis in infants) → convalescent; azithromycin 10 mg/kg (day 1, then 5 mg/kg days 2–5 if ≥6 months; 10 mg/kg daily for 5 days in infants <6 months) or erythromycin 12.5 mg/kg 6-hourly; treat contacts; DTP at 6, 10, 14 weeks.
10. Empyema: persistent fever on antibiotics, stony dull percussion, absent breath sounds → CXR/ultrasound, diagnostic tap, chest drain, cloxacillin + gentamicin (Staph) for 3–4 weeks.
11. Kerosene/paraffin ingestion: never induce vomiting or wash out the stomach; give oxygen, observe at least 6 h, CXR; antibiotics only if secondary infection.
12. Pneumonia killed about 740,000 under-fives in 2019 (WHO); pneumonia and diarrhoea together cause about 23% of under-five deaths (WHO 2024).

1 In a nutshell
2 Definitions & classification

3 Epidemiology & Uganda context
4 Pathophysiology
5 Clinical acumen
6 Clinical workup
7 Differential diagnosis
8 Investigations
9 Management
10 Complications
11 Ward-round & exam pearls
12 Mnemonics & memory aids
13 High-yield summary

# In a nutshell

**Lower respiratory tract infections (LRTIs)** are infections below the larynx: **pneumonia** (infection of the lung parenchyma), **bronchiolitis** (viral inflammation of the small airways in infants), **bronchitis**, and the special cases of **pertussis**, **empyema**, **aspiration** and **HIV-related pneumonia** (PCP). Pneumonia is the **single largest infectious cause of death in children worldwide** (WHO fact sheet), and in Uganda it is one of the top three killers of under-fives, alongside malaria and neonatal conditions.

The WHO/IMCI approach was designed so that a health worker with a watch and a pair of eyes can save most of these children:

1. **Count** the breaths for a full minute, **look** for chest indrawing, **check** for danger signs, and **measure $SpO_2$**.
2. **Classify**: no pneumonia (cough or cold) / **pneumonia** (oral amoxicillin at home) / **severe pneumonia** (refer, oxygen, injectable antibiotics).
3. **Give oxygen** when $SpO_2$ is below 90%. Hypoxaemia is the strongest predictor of death, and oxygen is the most life-saving drug on the ward.
4. **Review at 48 hours** and think of the reasons for failure: Staph, TB, HIV/PCP, empyema, foreign body, heart failure, malnutrition.

> **CLINICAL PEARL**
> The one thing never to forget: **put a pulse oximeter on every child with cough or difficult breathing.** Clinical signs miss about a third of hypoxaemic children, and a child with $SpO_2$ <90% is several times more likely to die. Oxygen is cheap; a missed reading is not.

# Definitions & classification

| Term | Meaning |
|---|---|
| **Pneumonia** | Infection of the lung parenchyma (alveoli). Clinically (WHO): cough or difficult breathing **plus fast breathing and/or chest indrawing** |
| **Fast breathing** (WHO/IMCI) | **<2 months: ≥60/min; 2–11 months: ≥50/min; 12–59 months: ≥40/min**; ≥5 years: >30/min (UCG 2023) |
| **Lower chest wall indrawing** | The lower chest wall goes **in** when the child breathes **in** (not just the soft tissue between the ribs); must be present all the time and in a calm child |
| **General danger signs** (IMCI) | **Unable to drink or breastfeed, vomits everything, convulsions (now or in this illness), lethargic or unconscious** |
| **Hypoxaemia** | $SpO_2$ **<90%** at sea level (<87% above 2,500 m) |
| **Bronchiolitis** | Acute viral LRTI in infants <2 years: coryzal prodrome, then cough, tachypnoea/recession and wheeze and/or crackles (NICE NG9) |
| **Empyema** | Pus in the pleural space |
| **Pneumatocoele** | Thin-walled air cyst in the lung, typical of **staphylococcal** pneumonia |

| Term | Meaning |
|---|---|
| **Aspiration pneumonitis/pneumonia** | Lung injury from inhaled gastric contents, hydrocarbons (kerosene) or oropharyngeal secretions; infection may follow |
| **Pertussis (whooping cough)** | Toxin-mediated respiratory infection with *Bordetella pertussis*: paroxysmal cough with whoop, vomiting, apnoea in infants |

## WHO/IMCI classification of cough or difficult breathing (2–59 months; UCG 2023; WHO 2024 guideline)

| Classification | Signs | Treatment |
|---|---|---|
| **Severe pneumonia (or very severe disease)** | Cough/difficult breathing + **any**: general danger sign; **central cyanosis or $SpO_2$ <90%**; **severe respiratory distress** (grunting, very severe chest indrawing, head nodding); (in UCG also: extrapulmonary features in malnourished or HIV-positive children) | **Admit/refer** after first dose of antibiotic; oxygen; **ampicillin (or benzylpenicillin) + gentamicin** |
| **Pneumonia** | **Fast breathing and/or chest indrawing**, no danger sign | **Oral amoxicillin 40 mg/kg twice daily for 5 days** (3 days for fast breathing alone in low-HIV areas); review in 2–3 days |
| **No pneumonia: cough or cold** | None of the above | Home care, soothing remedy, **no antibiotic**; cough >14 days → TB, asthma, pertussis |

UGANDA CONTEXT

The **WHO 2024 guideline on pneumonia and diarrhoea in children up to 10 years** confirmed that **chest-indrawing pneumonia without danger signs** is treated with **oral amoxicillin for 5 days as an outpatient** (rather than injectable antibiotics in hospital), and allowed **3 or 5 days** for fast-breathing pneumonia (3 days only in **low-HIV** settings). Uganda's adult HIV prevalence is about 5%, so **5 days** remains standard (UCG 2023). Uganda's **integrated community case management (iCCM)** lets VHTs give amoxicillin dispersible tablets for fast-breathing pneumonia in children 2–59 months.

## Young infants (<2 months)

UCG 2023: **consider every infant under 2 months with pneumonia as severe** (fast breathing ≥60, severe chest indrawing, grunting, not feeding, convulsions, drowsiness, apnoea, $SpO_2$ <90%) → admit, ampicillin + gentamicin. (WHO's 2015 possible serious bacterial infection guidance allows outpatient oral amoxicillin for **fast breathing alone in infants 7–59 days** when referral is impossible; know it exists, but in hospital practice we admit.)

## Pneumonia in children ≥5 years (UCG 2023)

- **Moderate**: fever, chest pain, cough, RR >30, no chest indrawing → oral amoxicillin.
- **Severe**: plus chest indrawing, pulse >120, temperature >39.5 °C, danger signs, $SpO_2$ <90% → admit, IV benzylpenicillin/ampicillin (± gentamicin) or ceftriaxone.

# Epidemiology & Uganda context

- **Global**: pneumonia accounted for **14% of under-five deaths (740,180 children in 2019)** and 22% of deaths in children aged 1–5 years (WHO fact sheet 2022). Pneumonia and diarrhoea together cause **23% of under-five deaths, about 1.17 million a year** (WHO 2024 guideline).
- **Causes** (UCG 2023; WHO): **neonates** – group B streptococcus, Klebsiella, *E. coli*, *S. aureus*, Chlamydia; **children <5 years** – *Streptococcus pneumoniae* (commonest bacterial), *Haemophilus influenzae*, *S. aureus*, viruses (**RSV** the commonest viral cause, influenza, measles, hMPV); **≥5 years** – pneumococcus, *Mycoplasma pneumoniae*, viruses; **HIV-infected infants** – *Pneumocystis jirovecii* (at least a quarter of pneumonia deaths in HIV-infected infants, WHO), CMV, TB, Gram negatives.
- **Risk factors**: malnutrition (wasting, stunting), HIV infection or exposure, no breastfeeding in the first 6 months, low birth weight, **indoor air pollution from charcoal/firewood**, crowding, parental smoking, missed vaccines (PCV, Hib, measles, pertussis), measles, sickle cell disease.
- **Prevention in Uganda**: exclusive breastfeeding to 6 months, **PCV and pentavalent (DTP-HepB-Hib) at 6, 10, 14 weeks, measles-rubella at 9 and 18 months** (UNEPI), cotrimoxazole prophylaxis for HIV-exposed infants, ART, improved cookstoves, handwashing, zinc and nutrition.
- **What is available**:

| Level | Usually available | Often missing |
|---|---|---|
| **VHT (iCCM)** | Respiratory timer, **amoxicillin DT**, referral | Oximeter (some now have) |
| **HC III** | Amoxicillin DT, first-dose injectable (benzylpenicillin/gentamicin), paracetamol | Oxygen, oximeter (improving), X-ray |
| **HC IV / general hospital** | Oxygen concentrators, pulse oximeters, ampicillin, gentamicin, ceftriaxone, cloxacillin, X-ray (some) | Reliable power for concentrators, blood culture, chest drains |
| **Regional / national referral** | X-ray, ultrasound, blood culture, chest drains, bubble CPAP (Mulago, some RRHs), HDU | PICU and ventilators (very few beds) |

**UGANDA CONTEXT**

Oxygen access has improved in Uganda through concentrators, cylinder plants at regional referral hospitals and **solar-powered oxygen** projects, but many wards still run out at night or during power cuts. Check the concentrator and the oximeter at the start of every shift, and **prioritise oxygen** for children with $SpO_2$ <90%. Bubble CPAP made from nasal prongs, tubing and a water bottle is used at Mulago and several regional hospitals for severe pneumonia and bronchiolitis.

# Pathophysiology

### How pneumonia makes a child hypoxic

1. Organisms reach the alveoli (aspiration of nasopharyngeal flora, inhaled droplets, or blood spread in Staph).
2. **Inflammation fills alveoli with fluid, neutrophils and fibrin** (consolidation).
3. **Ventilation–perfusion mismatch and shunt**: blood passes alveoli that are not ventilated → **hypoxaemia** (not corrected by breathing faster).
4. The child compensates by **breathing faster** (tachypnoea) and harder (**indrawing**, nasal flaring, grunting = breathing out against a partly closed glottis to keep alveoli open; head nodding = accessory neck muscles in an infant).
5. Fatigue, sepsis, dehydration, hypoglycaemia and **hypoxaemia** cause death; malnutrition and HIV blunt the signs (less fever, less tachypnoea).

### Why the signs work

- **Fast breathing** is the most sensitive single sign of pneumonia; age-specific cut-offs account for normal infant rates.
- **Chest indrawing** reflects stiff (non-compliant) lungs pulling on a compliant chest wall; it marks more severe disease.
- **Crackles** come from fluid in small airways and alveoli; **bronchial breathing** from consolidation transmitting tracheal sounds; but auscultation in a crying infant is unreliable, which is why WHO relies on counting and looking.

### Bronchiolitis

RSV infects the bronchiolar epithelium → **necrosis, oedema, mucus and cell debris plug the small airways** → air trapping (hyperinflation, wheeze) and atelectasis (crackles) → V/Q mismatch and hypoxaemia. Infants have tiny airways and are obligate nasal breathers, so a blocked nose worsens feeding and breathing. The illness **peaks on days 3–5** and cough resolves within 3 weeks in 90% (NICE NG9). $\beta_2$-agonists don't work because the problem is plugging, not smooth-muscle spasm.

### Pertussis

*B. pertussis* attaches to ciliated epithelium and releases **pertussis toxin** (lymphocytosis, hypoglycaemia, impaired immune response) and tracheal cytotoxin (ciliary damage). Thick mucus and damaged cilia cause **paroxysms** of coughing; the inspiratory **whoop** is air rushing through a narrowed glottis after the paroxysm. Young infants may not whoop: they **stop breathing (apnoea) and go blue**.

### Staphylococcal pneumonia and empyema

*S. aureus* produces toxins (e.g. Panton-Valentine leukocidin) that cause **necrosis**: abscesses, **pneumatocoeles**, **pneumothorax** and **empyema**. It often follows measles, influenza, skin sepsis or pyomyositis, and progresses fast.

### Hydrocarbon aspiration

Kerosene and paraffin have **low viscosity and surface tension**: tiny amounts spread over the airways, dissolve surfactant and cause chemical pneumonitis. Inducing vomiting gives the fluid a second chance to be aspirated.

## Clinical acumen

### Focused history

- **Cough** (duration: <14 days acute; ≥14 days think TB, asthma, pertussis, foreign body, HIV lung disease), **difficult or fast breathing**, fever, grunting, chest pain.
- **Danger signs**: able to drink or breastfeed? vomiting everything? convulsions? sleepy?
- **Paroxysms, whoop, post-tussive vomiting, apnoea or cyanosis** (pertussis); **sudden choking** (foreign body); **ingestion** of kerosene, paraffin or other chemicals.
- **Age** (≤2 months: neonatal sepsis organisms; <2 y: bronchiolitis), recent **measles**, skin boils or pyomyositis (Staph).
- **HIV**: mother's status, infant's PCR/antibody, cotrimoxazole, ART; **TB contact**; recurrent pneumonias.
- **Growth and nutrition**: weight-for-height Z-score, MUAC, oedema (SAM changes antibiotics and fluids), breastfeeding.
- **Immunisation**: PCV ×3, pentavalent ×3 (Hib, pertussis), MR ×2; BCG scar.
- **Treatment so far**: antibiotics (which, how long?), herbal medicines, "nebulisation" at a clinic.

- **Home**: cooking fuel, smokers, crowding, siblings ill.

### Examination

1. **Look** before you touch: alertness, colour (central cyanosis on the tongue/lips), nasal flaring, grunting, **head nodding**, lower chest wall indrawing, ability to drink.
2. **Count** the respiratory rate for **60 seconds** while the child is calm (on the mother's lap, not feeding, not crying; watch the abdomen in infants).
3. **$SpO_2$** with an appropriately sized probe (finger or toe in infants), good waveform.
4. **Temperature, HR, CRT, hydration, weight, MUAC, oedema.**
5. **Chest**: percussion (dull = consolidation; **stony dull** = effusion/empyema), breath sounds (reduced, bronchial), crackles, wheeze, tracheal position.
6. **Elsewhere**: oral thrush, lymphadenopathy, hepatosplenomegaly, parotid swelling (HIV); heart murmur, gallop, hepatomegaly (heart failure); skin sepsis; measles rash; BCG scar; signs of SAM.

### Red flags

RED FLAG

**Severe pneumonia, needs admission now:** any general danger sign (unable to drink/breastfeed, vomits everything, convulsions, lethargy/unconsciousness), **$SpO_2$ <90%** or central cyanosis, grunting, head nodding, very severe chest indrawing, RR ≥70, age <2 months, severe acute malnutrition, apnoea.

RED FLAG

**Pneumonia that is not responding**: still febrile and breathless after 48 h on correct antibiotics, rising RR, new dullness (empyema), sudden worsening (pneumothorax), pneumatocoeles (Staph), very low $SpO_2$ out of proportion to signs in an HIV-exposed infant (PCP).

### Thinking like a clinician

- **Count, don't guess.** "Fast breathing" by impression is wrong half the time. Use the respiratory timer and a full minute.
- **Fever + fast breathing in Uganda = pneumonia and/or malaria.** Do an RDT on every febrile child; **treat both** if both are present. Severe malaria with acidosis also causes deep, fast breathing.
- **Wheeze changes the story.** Fast breathing with wheeze in an infant → bronchiolitis or viral wheeze; in an older child → asthma (give salbutamol and **re-count**: if fast breathing resolves, it was bronchospasm, not pneumonia). WHO advises a rapid-acting bronchodilator trial before classifying wheezing children with fast breathing.
- **The thin, HIV-exposed infant with severe hypoxaemia and few chest signs** has PCP until proven otherwise: high-dose cotrimoxazole + steroid, alongside standard antibiotics.
- **Staph clues**: measles, skin boils/pyomyositis, rapid deterioration, pneumatocoeles, empyema, pneumothorax.
- **Persistent fever and cough >2 weeks with weight loss** → TB (see the TB topic); **sudden onset in a toddler** → foreign body.
- **Basal crackles + hepatomegaly + murmur** → heart failure (VSD, RHD, myocarditis): fluids and diuretics, not just antibiotics.
- **SAM + pneumonia**: signs are blunted; treat as complicated SAM (antibiotics for all, careful fluids, F-75, warmth, glucose).

# Clinical workup

## Step 0: First 5 minutes

1. **ETAT triage**: Airway and Breathing emergency signs (obstructed breathing, **central cyanosis, severe respiratory distress**), circulation (cold hands + CRT >3 s + weak fast pulse), coma/convulsions, severe dehydration. If present → resuscitation area.
2. **Oxygen** for $SpO_2$ <90% or any emergency sign (target **≥94% during resuscitation**, then ≥90%).
3. **Position**: sitting up on the mother's lap; clear the nose with gentle suction.
4. **Check glucose** (lethargic, convulsing or not feeding): treat hypoglycaemia <2.5 mmol/L (or <3 mmol/L in SAM) with **10% dextrose 5 mL/kg IV** (or 50 mL of 10% dextrose/sugar water by NG if no IV).
5. **Convulsions**: diazepam 0.5 mg/kg rectally or 0.2–0.3 mg/kg IV (UCG 2023); look for meningitis and malaria.
6. **Weigh** the child; **first dose of antibiotics** without delay in severe pneumonia (before referral if you are at a lower level).
7. **Shock** is uncommon in pneumonia; if present (and no SAM), give 10–20 mL/kg Ringer's lactate/normal saline and reassess; think sepsis and dehydration.

## Step 1: Focused history

- [ ] **Age** → <2 months = severe; <2 years with coryza and wheeze = bronchiolitis.
- [ ] **Duration of cough** → ≥14 days: TB, pertussis, asthma, foreign body, HIV lung disease.
- [ ] **Danger signs** (drink, vomit, convulsions, lethargy) → severe pneumonia.
- [ ] **Paroxysms, whoop, vomiting after cough, apnoea** → pertussis (macrolide, isolate, contacts).
- [ ] **Sudden choking or chemical ingestion** → foreign body, hydrocarbon pneumonitis.
- [ ] **HIV status of mother and child, cotrimoxazole, ART** → PCP and TB risk; ART decisions.
- [ ] **TB contact** → TB workup.
- [ ] **Measles, boils, pyomyositis** → Staph.
- [ ] **Antibiotics already taken** → treatment failure pathway.
- [ ] **Nutrition, oedema, breastfeeding** → SAM protocol.
- [ ] **Vaccines** (PCV, Hib, DTP, MR) → aetiology and catch-up.

## Step 2: Focused examination

**General**: alertness (AVPU), ability to drink, cyanosis, pallor, hydration, temperature, **weight, length/height, WHZ, MUAC, oedema**.

**Respiratory**: RR (60 s), $SpO_2$, **lower chest wall indrawing**, nasal flaring, grunting, head nodding, percussion, breath sounds (bronchial, reduced), crackles, wheeze, tracheal position.

**Other**: heart (murmur, gallop), liver size, ENT (otitis), mouth (thrush), nodes, skin (measles, boils), neck stiffness/fontanelle (meningitis).

| Document these positives | And these negatives |
|---|---|
| RR (counted for 60 s) and age cut-off | "No general danger signs" |
| Chest indrawing; grunting; head nodding | "No central cyanosis" |
| $SpO_2$ in air (and on oxygen) | "No wheeze" (or wheeze: re-count after bronchodilator) |
| Focal signs (dullness, bronchial breathing, crackles) | "No stony dullness; trachea central" |
| WHZ/MUAC, oedema, HIV status, vaccines | "No hepatomegaly or murmur" |

## Step 3: Bedside tests

| Test | How | What it tells you |
|---|---|---|
| **Respiratory rate** (timer) | 60 s, calm child | Fast breathing by age |
| **Pulse oximetry** | Right-sized probe; warm extremity; steady trace for 30–60 s | **<90% = severe pneumonia, give oxygen**; 90–93% = monitor closely |
| **Malaria RDT** | Every febrile child | Co-treat malaria |
| **Random blood glucose** | Lethargic, not feeding, SAM | Hypoglycaemia → 10% dextrose 5 mL/kg |
| **Haemoglobin (HemoCue)** | Pallor, severe illness | Severe anaemia (Hb <5 g/dL, or <7 with distress) → transfuse; heart failure from anaemia mimics pneumonia |
| **HIV rapid test / DNA PCR** | Status unknown; infants of positive mothers need PCR | PCP risk; ART; cotrimoxazole |
| **MUAC tape** | 6–59 months | <11.5 cm = SAM |
| **Bronchodilator trial** | Wheeze + fast breathing: salbutamol 2 puffs via spacer or 2.5 mg neb, re-count after 15–20 min | Fast breathing resolves = bronchospasm, not pneumonia |

## Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Chest X-ray** | Severe pneumonia, not responding at 48 h, suspected complications (effusion, empyema, pneumothorax, pneumatocoeles), suspected TB or foreign body. **Not needed** for non-severe pneumonia | Lobar consolidation (pneumococcus); patchy bronchopneumonia; **pneumatocoeles/empyema** (Staph); hilar nodes, miliary pattern (TB); diffuse interstitial "ground glass" (PCP); hyperinflation (bronchiolitis) | Staph → cloxacillin; empyema → drain; TB → Xpert, treat; PCP → cotrimoxazole. HC IV (some), all hospitals |
| **Chest ultrasound** | Suspected effusion/empyema | Fluid, septations, size | Guides tap and drain site. RRH/NRH, some HC IVs |
| **Full blood count** | Severe or complicated | Neutrophilia (bacterial); lymphocytosis >20 × $10^9$/L (pertussis); anaemia; low platelets | Supports diagnosis; transfusion |
| **Blood culture** | Severe pneumonia, before antibiotics if possible | Pneumococcus, Staph, Klebsiella, Salmonella | Tailors therapy. RRH/NRH only; yield low |
| **Pleural fluid** (diagnostic tap) | Effusion | Pus; pH <7.2, glucose low, LDH high (empyema); lymphocytes, high protein (TB); Gram stain, culture, **Xpert** | Chest drain for empyema; TB treatment |
| **GeneXpert** (sputum, gastric aspirate, **stool**, NPA) | Cough ≥2 weeks, TB contact, HIV, not responding | MTB detected ± rifampicin resistance | TB treatment |

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Urea, electrolytes** | Severe illness, IV fluids, SAM | Hyponatraemia (SIADH), AKI | Fluid restriction (about 2/3 maintenance if Na low) |
| **Nasopharyngeal swab PCR** (RSV, pertussis, influenza) | Research and outbreaks, private hospitals | Viral or pertussis diagnosis | Isolation, macrolide for pertussis. Rarely available |
| **Blood gas** | Severe distress, CPAP, HDU | Rising $pCO_2$ | CPAP/ventilation. NRH/ICU |

## Step 5: Putting it together

### Model one-line summary

> "A 10-month-old girl weighing 7.2 kg (WHZ −1.5, MUAC 12.4 cm), fully immunised for age including PCV ×3, HIV-exposed with a negative DNA PCR at 6 weeks and still breastfeeding (repeat test due), with 3 days of cough and fever, now unable to breastfeed, RR 64, lower chest wall indrawing, grunting and $SpO_2$ 86% in air, malaria RDT negative: **severe pneumonia with hypoxaemia**."

### Problem list

1. Severe pneumonia (danger sign + hypoxaemia).
2. Hypoxaemia: oxygen.
3. HIV-exposed infant, breastfeeding: repeat HIV test; consider PCP if hypoxaemia severe and test positive.
4. Moderate wasting risk (WHZ −1.5): monitor feeding.

**Working diagnosis**: *severe community-acquired pneumonia (likely pneumococcal) with hypoxaemia in an HIV-exposed infant.*

EXAM FAVOURITE

**Worked example 1: classify three children with cough.** (a) 8 months, RR 56, no indrawing, feeding well, $SpO_2$ 96% → fast breathing (≥50) → **pneumonia** → amoxicillin 40 mg/kg twice daily × 5 days at home, review in 2–3 days. (b) 3 years, RR 38, lower chest wall indrawing, drinking, $SpO_2$ 94% → indrawing → **pneumonia** (WHO 2014/2024: outpatient oral amoxicillin 5 days, unless other risks such as SAM or HIV infection; many Ugandan clinicians still admit). (c) 18 months, RR 52, not able to drink, $SpO_2$ 88% → **severe pneumonia** → admit, oxygen, ampicillin + gentamicin.

EXAM FAVOURITE

**Worked example 2: the 48-hour review.** A 2-year-old (11 kg) on ampicillin + gentamicin for 48 h is still febrile (39.5 °C), RR 60, $SpO_2$ 89% on 1 L/min, with new stony dullness and absent breath sounds over the right base. CXR: right pleural effusion with a pneumatocoele in the left upper zone. **Interpretation**: treatment failure from **staphylococcal pneumonia with empyema**. Action: ultrasound-guided **diagnostic tap** (thick pus) → **chest drain**; switch to **cloxacillin 50 mg/kg (550 mg) 6-hourly + gentamicin 7.5 mg/kg once daily** for at least 3 weeks; increase oxygen; blood culture; HIV test; nutritional support.

# Differential diagnosis

| Condition | Distinguishing features | Key investigation |
|---|---|---|
| **Pneumonia** | Fever, cough, fast breathing, indrawing, focal crackles/bronchial breathing | Clinical; $SpO_2$; CXR if severe |
| **Bronchiolitis** | <2 y, coryza then cough, fine crackles and wheeze, hyperinflation, little fever | Clinical |
| **Asthma / viral wheeze** | Recurrent wheeze, responds to bronchodilator, atopy | Bronchodilator trial |
| **Malaria** (severe, with acidosis) | Deep (acidotic) breathing, fever, pallor, prostration, clear chest | RDT/blood smear, lactate |
| **Heart failure** (VSD, PDA, RHD, cardiomyopathy, severe anaemia) | Hepatomegaly, gallop, murmur, poor feeding/sweating, oedema | CXR (cardiomegaly), Hb, echo |
| **Pulmonary TB** | Cough ≥2 weeks, weight loss, TB contact, non-response to antibiotics | Xpert, CXR, TST |
| **PCP (HIV)** | HIV-exposed/infected infant 2–6 months, severe hypoxaemia, few crackles, high LDH | HIV PCR/test; CXR interstitial |
| **Foreign body aspiration** | Sudden choking, unilateral signs | Inspiratory/expiratory CXR; bronchoscopy |
| **Pertussis** | Paroxysms, whoop, vomiting, apnoea; clear chest between bouts; lymphocytosis | Clinical; PCR (rare) |
| **Hydrocarbon aspiration** | Kerosene smell, cough after ingestion | History; CXR at 6 h |
| **Metabolic acidosis** (DKA, severe dehydration, salicylate) | Deep sighing breathing, clear chest | Glucose, ketones, gas |
| **Sepsis / meningitis** | Fast breathing from fever and acidosis; neck stiffness, bulging fontanelle | LP, blood culture |

# Investigations

| Test | What you are looking for | Why |
|---|---|---|
| $SpO_2$ | <90% | Oxygen; severe pneumonia |
| Malaria RDT | Positive | Co-infection is common |
| CXR (selected) | Consolidation, effusion, pneumatocoeles, pneumothorax, TB, PCP pattern | Complications; failure; alternative diagnoses |
| FBC | Neutrophilia, lymphocytosis, anaemia | Bacterial vs pertussis; transfusion |
| Blood culture | Organism | Targeted therapy |
| Pleural tap | Pus, pH, Xpert | Empyema vs TB |
| HIV test | Positive | PCP, ART |
| Xpert MTB/RIF Ultra | TB | Chronic or non-responding pneumonia |
| Glucose, electrolytes | Hypoglycaemia, hyponatraemia | Supportive care |

# Management

## A. Pneumonia (non-severe): outpatient

1. **Oral amoxicillin dispersible tablets (250 mg) 40 mg/kg per dose twice daily for 5 days** (UCG 2023; WHO 2024 minimum 80 mg/kg/day):
   - 2–12 months (4–<10 kg): **250 mg (1 tab)** twice daily
   - 1–3 years (10–<14 kg): **500 mg (2 tabs)** twice daily
   - 3–5 years (14–19 kg): **750 mg (3 tabs)** twice daily
2. **Wheeze**: salbutamol inhaler with spacer 1–2 puffs (or more) 4–6-hourly until wheeze stops (UCG 2023).
3. **Fever**: paracetamol 15 mg/kg 6-hourly.
4. **Home care**: continue breastfeeding and feeding, extra fluids, clear the nose, keep warm (not overwrapped).
5. **Return immediately** if unable to drink/breastfeed, breathing faster or harder, becomes sicker, convulsions.
6. **Review after 2–3 days** (IMCI/UCG): better → complete course; same or worse → severe pathway (refer, injectable antibiotics, oxygen).

## B. Severe pneumonia: inpatient (UCG 2023, WHO Pocket Book 2013)

1. **Admit**; refer from HC III/IV after the first dose of antibiotics (IM if no IV access).
2. **Oxygen** if $SpO_2$ <90% (see section E).
3. **Antibiotics (first line)**: **ampicillin 50 mg/kg IV/IM 6-hourly** (or **benzylpenicillin 50,000 IU/kg IV/IM 6-hourly**) **plus gentamicin 7.5 mg/kg IV/IM once daily**, for **at least 5 days** (up to 10). Switch to oral amoxicillin 40 mg/kg twice daily when improving, to complete at least 5 days in total (UCG) (WHO: total 5–10 days).
4. **Second line** (no improvement after **48 h**, or very severe): **ceftriaxone 80 mg/kg IV/IM once daily** (UCG 2023; max 2 g), 7–10 days.
5. **Suspected Staph** (pneumatocoeles, empyema, skin sepsis, after measles, rapid deterioration): **cloxacillin 50 mg/kg IV/IM 6-hourly + gentamicin 7.5 mg/kg once daily**, for **at least 3 weeks** (UCG: 21 days).
6. **Last resort** if the above are unavailable: chloramphenicol 25 mg/kg IV 6-hourly (not in neonates <7 days or preterm).
7. **Supportive care**:
   - **Fluids**: encourage breastfeeding/oral fluids; NG feeds or IV fluids if unable to drink. Give **maintenance only** (Holliday–Segar), and consider restricting to about two-thirds of maintenance if very severe or hyponatraemic (SIADH). Avoid overload, especially in small and malnourished children (UCG).
   - **Fever**: paracetamol 15 mg/kg 6-hourly (UCG 10 mg/kg 4–6-hourly).
   - **Wheeze**: salbutamol 1–2 puffs (or more via spacer) 4–6-hourly.
   - **Gentle suction** of thick upper-airway secretions; **no chest physiotherapy** in acute pneumonia.
   - **Convulsions**: diazepam 0.5 mg/kg rectally or 0.2–0.3 mg/kg IV; if continuous, phenobarbital 15–20 mg/kg IV/IM loading (UCG lists 10–15 mg/kg IM), then 3–5 mg/kg/day.
   - **Glucose**: prevent hypoglycaemia (feeds; dextrose if not feeding).
8. **Monitoring** (UCG 2023): RR every 2 hours (more often if very sick), temperature 6-hourly, **$SpO_2$ at least every 3–6 hours** (UCG minimum 12-hourly; every 3 h while on oxygen is better practice), feeding, alertness, work of breathing.
9. **Treat co-existing conditions**: malaria (IV artesunate if severe), SAM (WHO 2023 SAM protocol), anaemia (transfuse if Hb <5 g/dL or <7 with distress), HIV (cotrimoxazole, ART).

## C. Young infants (<2 months)

- **Admit all** (UCG 2023). Keep warm (skin-to-skin), breastfeed or expressed milk by NGT, prevent hypoglycaemia.

- **Ampicillin 50 mg/kg IV 6-hourly** (12-hourly in the first week of life) **+ gentamicin 5 mg/kg once daily** (term, first week of life) / **7.5 mg/kg once daily** (>7 days); preterm/LBW: reduce dose (specialist).
- Severely ill: **ceftriaxone 100 mg/kg once daily** (UCG 2023) (avoid with calcium-containing IV fluids and in jaundiced neonates; cefotaxime preferred where available).
- Oxygen to keep $SpO_2$ **≥90%** (UCG: >94% for neonates with emergency signs).
- At least 5 days, and 3 days after the infant is well; **10 days** if septicaemia; **21 days** if meningitis (UCG).

### D. Drug table: LRTI

| Drug | Indication | Dose | Route | Frequency | Duration | Max / notes |
|---|---|---|---|---|---|---|
| **Amoxicillin DT** | Pneumonia (non-severe); step-down | **40 mg/kg** | Oral | **12-hourly** | **5 days** (3 days fast breathing only in low-HIV settings) | 1 g/dose |
| **Ampicillin** | Severe pneumonia, infants | **50 mg/kg** | IV/IM | **6-hourly** (12-hourly first week of life) | ≥5 days | 2 g/dose |
| **Benzylpenicillin** | Severe pneumonia (alternative) | **50,000 IU/kg** | IV/IM | 6-hourly | ≥5 days | 2–4 MU/dose |
| **Gentamicin** | Severe pneumonia (with ampicillin) | **7.5 mg/kg** (>1 week old); **5 mg/kg** (first week) | IV/IM | **Once daily** | ≥5 days | Avoid prolonged courses; renal and ototoxicity |
| **Ceftriaxone** | Second line; severely ill infants | **80 mg/kg** (100 mg/kg in severely ill infants, UCG) | IV/IM | Once daily | 7–10 days | 2 g (4 g meningitis); not with IV calcium in neonates |
| **Cloxacillin** | Staph pneumonia, empyema | **50 mg/kg** (25–50) | IV/IM then oral | **6-hourly** | **≥21 days** | 2 g/dose |
| **Chloramphenicol** | If others unavailable | **25 mg/kg** | IV | 6-hourly | 5–10 days | Not <7 days old or preterm (grey baby) |
| **Cotrimoxazole (high dose)** | PCP | **120 mg/kg/day** of combined drug (= TMP 20 mg/kg/day) in 3–4 divided doses (UCG); WHO: TMP 8 mg/kg + SMX 40 mg/kg 8-hourly | Oral/IV | 6–8-hourly | **21 days**, then prophylaxis | Rash, marrow suppression |
| **Prednisolone** | PCP with hypoxaemia | **2 mg/kg/day** (in divided doses) for 5 days then taper over 21 days (UCG; WHO 1–2 mg/kg/day for 1 week) | Oral | – | ~21 days total | – |
| **Azithromycin** | Pertussis | **<6 months: 10 mg/kg once daily × 5 days; ≥6 months: 10 mg/kg day 1 (max 500 mg), then 5 mg/kg days 2–5 (max 250 mg)** | Oral | Once daily | 5 days | Preferred in neonates (CDC); fewer GI effects |
| **Erythromycin** | Pertussis (UCG first line), Mycoplasma | **12.5 mg/kg** (UCG 10–15 mg/kg) | Oral | **6-hourly** | **7 days** (UCG) to 14 days | Avoid in <1 month (pyloric stenosis) |

| Drug | Indication | Dose | Route | Frequency | Duration | Max / notes |
|---|---|---|---|---|---|---|
| **Doxycycline** | Mycoplasma in **>8 y** | 2 mg/kg (max 100 mg) | Oral | 12-hourly | 7–10 days | Not <8 y |
| **Metronidazole** | Aspiration pneumonia, lung abscess | 7.5 mg/kg (UCG 12.5 mg/kg IV for lung abscess) | IV/oral | 8-hourly | With penicillin; lung abscess 4–6 weeks | – |
| **Paracetamol** | Fever, pain | 15 mg/kg | Oral | 6-hourly | Prn | 60 mg/kg/day |
| **Salbutamol** | Wheeze | 100 µg/puff, 2 puffs (or more) via spacer; or 2.5 mg neb | Inhaled | 4–6-hourly | Until wheeze stops | – |

DRUG DOSING

**Gentamicin once daily** is standard: 7.5 mg/kg for children over 1 week old (UCG 2023), 5 mg/kg in the first week of life. Write the actual mg dose using today's weight, and avoid courses longer than 7–10 days without levels.

## E. Hypoxaemia and oxygen therapy (WHO 2016 oxygen therapy for children; UCG 2023)

### Who gets oxygen

- **$SpO_2$ <90%** (at Ugandan altitudes; <87% above 2,500 m) → oxygen.
- **Emergency signs** (obstructed/absent breathing, severe respiratory distress, central cyanosis, shock, coma, convulsions) → oxygen during resuscitation to **$SpO_2$ ≥94%**.
- **No oximeter**: give oxygen if **central cyanosis, grunting, nasal flaring, head nodding, inability to drink/feed because of breathing difficulty, severe lower chest wall indrawing, RR ≥70, or depressed consciousness** (WHO 2016; WHO 2024 adds head nodding, nasal flaring, grunting or RR ≥20 above the age cut-off as signs of hypoxaemia).

### Delivery

| Device | Flow | $FiO_2$ approx | Notes |
|---|---|---|---|
| **Nasal prongs** (preferred) | **0.5–1 L/min neonates; 1–2 L/min infants; 2–4 L/min older children** | 30–35% (at 1–2 L/min in infants) | Comfortable; child can feed; no humidification needed at these flows |
| **Nasal catheter** (8 Fr, inserted to the distance nose–inner eyebrow) | 0.5–2 L/min | 35–40% | Needs humidification if high flow; risk of gastric distension |
| **Face mask** | 5–10 L/min (≥5 to avoid $CO_2$ rebreathing) | 40–60% | Older children, emergencies |
| **Head box** | 2–3 L/min/kg | Variable | Neonates; wasteful, $CO_2$ build-up |
| **Bubble CPAP** | 5–10 cm $H_2O$ via nasal prongs | – | Severe pneumonia/bronchiolitis not improving on standard oxygen (RRH/NRH) |

### Monitoring and weaning

- Check $SpO_2$ at least every **3 hours** on oxygen; check prongs are in the nose, tubing unkinked, concentrator working.

- **Trial off oxygen daily** when stable: if $SpO_2$ stays **≥90% in room air for 10–15 minutes** (and the child is not distressed), stop and recheck after 1 h and then 3-hourly for the first day.
- Concentrators need constant power: keep a cylinder as back-up; use one concentrator with a flow splitter to treat several children only if each child's flow is measured.

> **UGANDA CONTEXT**
> Uganda's 2023 guideline (UCG) says oxygen when **$SpO_2$ <90%** for children with pneumonia, and **>94%** for young infants with emergency signs. NICE uses 92% (bronchiolitis). For exams in Uganda: **<90% = give oxygen**.

### F. Bronchiolitis (UCG 2023 5.2.1; NICE NG9)

1. **Diagnose**: <2 years (usually <1), coryza 1–3 days, then cough, fast breathing or recession, and wheeze and/or crackles. Worst on days 3–5.
2. **Assess severity**: severe if **age <3 months**, RR >60–70, difficulty feeding (<50–75% of usual), **$SpO_2$ <92%**, apnoea, grunting, cyanosis, exhaustion (UCG; NICE).
3. **Mild–moderate** (feeding, no hypoxaemia): home care, saline drops, small frequent feeds, paracetamol for fever, return if worse.
4. **Admit if severe**:
   - **Oxygen** to keep $SpO_2$ **≥90%** (WHO) / ≥92% (NICE): nasal prongs 1–2 L/min (UCG: humidified 1–2 L/min).
   - **Feeding**: small frequent breastfeeds; **NG/OG feeds** if not feeding adequately; IV fluids only if NG not tolerated (isotonic, about maintenance).
   - **Gentle nasal suction** if secretions block the nose.
   - **CPAP** if impending respiratory failure.
5. **What not to give** (NICE NG9): antibiotics (unless secondary bacterial infection), salbutamol, adrenaline, hypertonic saline, steroids, ipratropium, montelukast, chest physiotherapy.

> **UGANDA CONTEXT**
> **UCG 2023 differs from NICE**: for severe bronchiolitis UCG suggests a **trial of salbutamol** (2 puffs via spacer every 30 min or 2.5 mg nebulised; **continue 6-hourly only if it helps, stop if no response**) and **nebulised adrenaline 1:1000, 1 mL in 2–4 mL saline, every 2–4 h**; it also lists total fluids of **150 mL/kg/24 h** plus losses. NICE NG9 and WHO recommend **no bronchodilators** and **no more than maintenance fluids** (overload and SIADH risk). On Kampala wards most consultants follow the supportive (NICE/WHO) approach and allow a single salbutamol trial in older infants with wheeze. Check your unit's practice.

**Discharge**: $SpO_2$ ≥90–92% in air for 4 h (including sleep), feeding ≥75% of normal, clinically stable; counsel that cough may last 3 weeks.

### G. Pertussis (UCG 2023 5.2.8)

1. **Recognise**: catarrhal stage (1–2 weeks, most infectious), **paroxysmal stage** (1–6 weeks: bouts of coughing ending in a **whoop**, **post-tussive vomiting**, subconjunctival haemorrhages; infants <6 months: **apnoea and cyanosis**, may not whoop), convalescent stage (weeks to months, "100-day cough").
2. **Admit** infants <6 months, any with apnoea, cyanosis, pneumonia, convulsions, dehydration or malnutrition.
3. **Antibiotics** (within 3 weeks of cough onset, UCG): **azithromycin** (preferred, especially <1 month) or **erythromycin 12.5 mg/kg 6-hourly** (UCG 10–15 mg/kg) for 7 days (UCG; 14 days classically). Reduces infectiousness; changes the course only if early.
4. **Supportive**: oxygen for cyanosis/apnoea ($SpO_2$ <90%), **gentle suction** of secretions after paroxysms, **small frequent feeds** (re-feed after vomiting), NG feeds if needed, monitor for apnoea (put near the nursing

station).

5. **Do NOT give**: cough mixtures, sedatives, mucolytics, antihistamines (UCG: "useless in pertussis").
6. **Isolate** until 5 days of antibiotics.
7. **Contacts**: macrolide prophylaxis for household contacts (especially infants and pregnant women); complete DTP vaccination; DTP at 6, 10, 14 weeks (UNEPI).
8. **Complications**: pneumonia (new fever), apnoea, convulsions, encephalopathy, malnutrition, dehydration, hernias, rectal prolapse, subconjunctival haemorrhage. Notifiable.

### H. Empyema and parapneumonic effusion

1. Suspect: **persistent fever** after 48–72 h of antibiotics, **stony dull percussion**, reduced/absent breath sounds, mediastinal shift.
2. **CXR + ultrasound**; **diagnostic tap** (pus, Gram stain, culture, **Xpert**, pH, glucose).
3. **Chest drain** (intercostal, 5th intercostal space mid-axillary line, underwater seal) for frank pus or large/complicated effusion; surgery (decortication) at NRH for loculated empyema not draining.
4. **Antibiotics**: **cloxacillin 50 mg/kg 6-hourly + gentamicin 7.5 mg/kg once daily** (Staph cover), or ceftriaxone if pneumococcal; continue **3–4 weeks** (IV until afebrile, then oral).
5. **Exclude TB** (lymphocytic exudate, TB contact, HIV).

### I. HIV-related pneumonia

- **HIV-exposed or infected infant with severe pneumonia**: standard antibiotics **plus high-dose cotrimoxazole for PCP** (especially age 2–6 months, severe hypoxaemia, few chest signs) for **21 days**, **prednisolone** if hypoxaemic, then cotrimoxazole prophylaxis. Test (DNA PCR <18 months), and start **ART** within 2 weeks (not delayed by PCP; TB requires attention to timing).
- Consider **TB**, CMV, LIP, bronchiectasis in HIV-positive children with chronic or recurrent pneumonia.

### J. Aspiration

| Situation | Management |
|---|---|
| **Kerosene/paraffin/hydrocarbon ingestion** | **Do not induce vomiting, no gastric lavage, no activated charcoal**. Remove contaminated clothes, wash skin. Oxygen if $SpO_2$ <90%. **Observe ≥6 hours**; CXR at 6 h if symptomatic. Antibiotics only if secondary bacterial pneumonia (fever after 24–48 h with new signs). |
| **Foreign body** (groundnut, bean, maize, bead) | Choking + effective cough → encourage coughing. Ineffective cough: infant – 5 back blows + 5 chest thrusts; child >1 y – 5 back blows + 5 abdominal thrusts; unconscious → CPR. Later presentation (unilateral wheeze, recurrent pneumonia in one lobe) → inspiratory/expiratory CXR, **rigid bronchoscopy** (Mulago ENT/surgery). |
| **Aspiration pneumonia** (cerebral palsy, reduced consciousness, swallowing problems) | Ampicillin + gentamicin + **metronidazole**, or amoxicillin-clavulanate; positioning, feeding assessment, NG feeding. |

### K. Discharge and prevention

- Discharge when off oxygen ≥24 h, feeding well, afebrile, improving; complete antibiotics at home.
- **Before discharge**: check and **catch up vaccines** (PCV, pentavalent, MR), HIV status, nutrition (enrol in OTC/ITC if SAM/MAM), counsel on smoke, breastfeeding, handwashing, danger signs; follow up in 1–2 weeks.

### Setting-specific plan

| Setting | What to do |
|---|---|
| **VHT (iCCM)** | Count RR with a timer; fast breathing (2–59 months) → amoxicillin DT 5 days; any danger sign or chest indrawing → first dose amoxicillin and refer |
| **HC III** | Classify; pneumonia → amoxicillin 5 days; severe → first dose IM ampicillin/benzylpenicillin + gentamicin, oxygen if available, refer with a referral note |
| **HC IV / general hospital** | Admit severe pneumonia; oxygen; IV/IM ampicillin + gentamicin; ceftriaxone if failing; CXR if available; refer for empyema/CPAP |
| **RRH / NRH** | All above, chest drains, CPAP, blood cultures, Xpert, HDU, bronchoscopy (NRH) |

## Complications

| Complication | Clues | Action |
|---|---|---|
| **Hypoxaemia / respiratory failure** | $SpO_2$ <90%, exhaustion | Oxygen, CPAP, ICU |
| **Parapneumonic effusion / empyema** | Persistent fever, stony dullness | Tap, drain, prolonged antibiotics |
| **Lung abscess / pneumatocoeles** | Staph, cavities | Cloxacillin; most pneumatocoeles resolve |
| **Pneumothorax / pyopneumothorax** | Sudden deterioration, hyper-resonance | Needle decompression, chest drain |
| **Sepsis, meningitis** | Shock, neck stiffness, bulging fontanelle | LP, meningitic doses |
| **Hyponatraemia (SIADH)** | Seizures, lethargy | Fluid restriction, sodium correction |
| **Heart failure** | Hepatomegaly, gallop | Fluid restriction, furosemide 1 mg/kg |
| **Malnutrition, growth faltering** | Weight loss | Nutritional rehabilitation |
| **Long-term** | Bronchiectasis (after measles, pertussis, recurrent pneumonia), post-infectious bronchiolitis obliterans | Physiotherapy, specialist |

## Ward-round & exam pearls

**Model case presentation (child):**

> "Namubiru is a 10-month-old girl from Nansana, weighing 7.2 kg (WHZ –1.5, MUAC 12.4 cm, no oedema), up to date with UNEPI vaccines for age including PCV and pentavalent ×3 and MR1 at 9 months, HIV-exposed with a negative DNA PCR at 6 weeks, still breastfeeding plus family foods. She presented with 3 days of cough and fever and was unable to breastfeed today. On arrival: lethargic but rousable, T 39.0 °C, RR 64 (counted for 60 s), HR 168, $SpO_2$ 86% in air, lower chest wall indrawing, grunting, bronchial breathing in the right upper zone; no wheeze, no hepatomegaly; malaria RDT negative; RBS 4.8 mmol/L. Classification: **severe pneumonia with hypoxaemia**. She is on oxygen 1 L/min via nasal prongs ($SpO_2$ 95%), IV ampicillin 360 mg 6-hourly and gentamicin 54 mg once daily, NG expressed breast milk, paracetamol. Plan: monitor RR, $SpO_2$ 3-hourly; repeat HIV test (rapid antibody at 9 months+, PCR if positive); review at 48 h for failure; CXR if not improving; nutritional follow-up."

### Likely viva questions with answers

1. *What are the WHO fast-breathing cut-offs?* <2 months ≥60; 2–11 months ≥50; 12–59 months ≥40 breaths/min.
2. *Name the general danger signs.* Unable to drink/breastfeed, vomits everything, convulsions, lethargic or unconscious.
3. *What is the first-line treatment of severe pneumonia?* Oxygen if $SpO_2$ <90%, ampicillin 50 mg/kg 6-hourly (or benzylpenicillin 50,000 IU/kg) + gentamicin 7.5 mg/kg once daily, at least 5 days.
4. *Which organisms cause pneumatocoeles? Staphylococcus aureus* (also Klebsiella); treat with cloxacillin + gentamicin for 3 weeks.
5. *When do you suspect PCP?* HIV-exposed/infected infant 2–6 months, severe hypoxaemia with few signs, not on cotrimoxazole prophylaxis.
6. *How do you wean oxygen?* Daily trial off when stable; stop if $SpO_2$ ≥90% in air for 10–15 min, then recheck.
7. *What is the treatment of bronchiolitis?* Supportive: oxygen, feeds (NG), nasal suction; no routine antibiotics or steroids.

### OSCE tips

- **Counting RR**: expose the chest, child calm (on mother's lap, not feeding), watch the abdomen in infants, count a **full minute** with a timer, state the age cut-off out loud.
- **Chest indrawing**: say "the lower chest wall goes in when the child breathes in" and check the child is calm (not crying, not feeding).
- **Oxygen station**: choose prongs size, set the flow by age, fix prongs with tape, check $SpO_2$ after 15 min, explain weaning.
- **Counselling a mother on amoxicillin**: dose with the tablet in breast milk or water, twice daily for 5 days even if better, return signs, follow-up in 2–3 days.

## Mnemonics & memory aids

- **Fast breathing "60–50–40"**: <2 months 60, 2–11 months 50, 1–5 years 40.
- **Danger signs "DVCL"**: can't **D**rink, **V**omits everything, **C**onvulsions, **L**ethargic/unconscious.
- **Hypoxaemia without an oximeter "CHIN-GRuNt"**: **C**yanosis, **H**ead nodding, **I**nability to feed, **N**asal flaring, **G**runting, **R**ate ≥70 (severe tachypnoea), u**N**conscious/lethargic.
- **Why is pneumonia not getting better? "STEP-HIV-F"**: **S**taph, **T**B, **E**mpyema, **P**CP, **H**eart failure, **I**nadequate dose/adherence, **V**irus or wrong diagnosis (bronchiolitis/asthma), **F**oreign body.
- **Pertussis stages "CPC"**: **C**atarrhal, **P**aroxysmal, **C**onvalescent.

## High-yield summary

### Numbers & doses to know

| Item | Value |
|---|---|
| Fast breathing | **≥60** (<2 m), **≥50** (2–11 m), **≥40** (1–4 y), **>30** (≥5 y, UCG) |
| Hypoxaemia | $SpO_2$ **<90%** (oxygen); target **≥94%** with emergency signs |
| Severe tachypnoea | RR **≥70** |
| Amoxicillin (pneumonia) | **40 mg/kg 12-hourly × 5 days** (3 days fast breathing only, low-HIV areas) |
| Amoxicillin by age (250 mg DT) | 2–12 m: **1 tab**; 1–3 y: **2 tabs**; 3–5 y: **3 tabs** twice daily |

| Item | Value |
|---|---|
| Ampicillin | **50 mg/kg 6-hourly** IV/IM |
| Benzylpenicillin | **50,000 IU/kg 6-hourly** |
| Gentamicin | **7.5 mg/kg once daily** (5 mg/kg first week of life) |
| Ceftriaxone | **80 mg/kg once daily** (100 mg/kg sick infants) |
| Cloxacillin (Staph) | **50 mg/kg 6-hourly × ≥21 days** |
| Cotrimoxazole for PCP | **120 mg/kg/day** (TMP 20 mg/kg/day) in 3–4 doses × **21 days** |
| Erythromycin (pertussis) | **12.5 mg/kg 6-hourly** × 7 days (UCG; 14 classically) |
| Nasal prong flow | **0.5–1 L/min** neonates; **1–2 L/min** infants; **2–4 L/min** older |
| Wean oxygen | $SpO_2$ **≥90%** in air for **10–15 min** |
| Review | Outpatient pneumonia at **2–3 days**; inpatient failure at **48 h** |
| Hydrocarbon ingestion | Observe **≥6 h** |

## Classic exam traps

- **"Crackles are needed to diagnose pneumonia."** No: WHO diagnosis is fast breathing and/or chest indrawing; auscultation is unreliable in infants.
- **"Chest indrawing = admit for IV antibiotics."** Since WHO 2014 (confirmed 2024), indrawing **without danger signs** can be treated with oral amoxicillin at home; admit if danger signs, $SpO_2$ <90%, SAM, age <2 months, or HIV-infected/unable to follow up.
- **"Cotrimoxazole for pneumonia."** No: amoxicillin is first line; cotrimoxazole is for PCP and prophylaxis.
- **"Salbutamol, steroids and antibiotics for bronchiolitis."** International guidance: supportive care only.
- **"Induce vomiting after kerosene ingestion."** Never.
- **"Ceftriaxone first for every admitted child."** First line is ampicillin/penicillin + gentamicin; ceftriaxone is second line (AWaRe Watch antibiotic).
- **"Normal $SpO_2$ means mild."** A child with danger signs is severe whatever the $SpO_2$.
- **"Fever + fast breathing = pneumonia, no need for an RDT."** Always test for malaria in Uganda.

## Questions seniors ask

**How do you classify this child?** State the age, RR against the cut-off, chest indrawing, danger signs and $SpO_2$, then the WHO class: no pneumonia, pneumonia, or severe pneumonia.

**Why ampicillin and gentamicin, not ceftriaxone, first?** They cover pneumococcus, H. influenzae and Gram negatives effectively, are Access antibiotics, cheaper and reduce resistance; ceftriaxone is reserved for failure or very sick infants.

**The child is not better at 48 hours. What do you think about?** Staph (pneumatocoeles, empyema), wrong diagnosis (TB, PCP, heart failure, foreign body), complications (effusion, abscess), malaria/anaemia/SAM, poor adherence or dosing; do a CXR, HIV test, Xpert and switch to ceftriaxone or cloxacillin as appropriate.

**When do you stop oxygen?** When the child is clinically stable and $SpO_2$ stays ≥90% in room air for 10–15 minutes on a daily trial off; recheck after an hour and overnight.

**How would you manage pertussis in a 6-week-old?** Admit, oxygen and suction for apnoea/cyanosis, small frequent or NG feeds, azithromycin 10 mg/kg once daily for 5 days, isolate, treat household contacts, complete DTP doses.

**How does HIV change your management of severe pneumonia?** Add high-dose cotrimoxazole for PCP (21 days) with prednisolone if hypoxaemic in infants, consider TB, start/optimise ART, and continue cotrimoxazole prophylaxis.

**What prevents pneumonia in Uganda?** Exclusive breastfeeding, PCV/Hib/pertussis/measles vaccines, good nutrition and zinc, clean cooking fuel, handwashing, HIV prevention and early ART, cotrimoxazole for HIV-exposed infants.

SELF-TEST

# Flashcards

27 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | WHO/IMCI fast-breathing cut-offs by age. | **<2 months ≥60/min; 2–11 months ≥50/min; 12–59 months ≥40/min**; ≥5 years >30/min (UCG 2023). Count for a full 60 s in a calm child. |
| 2 | IMCI general danger signs. | **Unable to drink or breastfeed, vomits everything, convulsions, lethargic or unconscious.** Any one = severe disease → refer/admit. |
| 3 | WHO classification: 'Pneumonia' vs 'Severe pneumonia' (2–59 months). | **Pneumonia**: cough/DIB + fast breathing and/or chest indrawing, no danger sign → oral amoxicillin. **Severe**: + any danger sign, central cyanosis/**$SpO_2$ <90%**, or severe respiratory distress (grunting, very severe indrawing) → admit, oxygen, injectable antibiotics. |
| 4 | Amoxicillin for non-severe pneumonia: dose and duration (UCG 2023/WHO 2024). | **40 mg/kg twice daily (80 mg/kg/day) for 5 days.** WHO 2024: 3 days acceptable for fast breathing only in **low-HIV** areas; Uganda uses 5. DT 250 mg: 2–12 m 1 tab, 1–3 y 2 tabs, 3–5 y 3 tabs, twice daily. |
| 5 | First-line antibiotics for severe pneumonia (UCG 2023). | **Ampicillin 50 mg/kg IV/IM 6-hourly** (or **benzylpenicillin 50,000 IU/kg 6-hourly**) **+ gentamicin 7.5 mg/kg once daily**, at least 5 days (up to 10), then oral amoxicillin. |
| 6 | Second-line antibiotic if severe pneumonia is not better after 48 h. | **Ceftriaxone 80 mg/kg IV/IM once daily** (UCG 2023). Also look for Staph, empyema, TB, PCP, heart failure, foreign body. |
| 7 | Treatment of suspected staphylococcal pneumonia. | **Cloxacillin 50 mg/kg (25–50) IV/IM 6-hourly + gentamicin 7.5 mg/kg once daily** for **at least 21 days** (UCG). Clues: pneumatocoeles, empyema, pneumothorax, skin sepsis, after measles. |
| 8 | How are infants <2 months with pneumonia managed (UCG 2023)? | All treated as **severe**: admit, keep warm, breast milk/NGT, oxygen; **ampicillin 50 mg/kg 6-hourly + gentamicin 5 mg/kg (first week) or 7.5 mg/kg once daily**; ceftriaxone 100 mg/kg if severely ill; ≥5 days (10 if sepsis, 21 if meningitis). |
| 9 | When is oxygen given (WHO/UCG) and what is the target? | **$SpO_2$ <90%** → oxygen to keep ≥90%. During resuscitation with emergency signs, target **≥94%**. |
| 10 | Signs predicting hypoxaemia when no oximeter is available. | **Central cyanosis, grunting, head nodding, nasal flaring, inability to drink/feed due to breathing, severe chest indrawing, RR ≥70, depressed consciousness** (WHO 2016; WHO 2024: RR ≥20 above the age cut-off). |
| 11 | Nasal prong oxygen flow rates by age. | **Neonates 0.5–1 L/min; infants 1–2 L/min; older children 2–4 L/min.** Face mask needs ≥5 L/min. Check $SpO_2$ 3-hourly; wean when $SpO_2$ ≥90% in air for 10–15 min. |
| 12 | Define bronchiolitis (NICE NG9). | Child <2 years (peak 3–6 months) with **coryzal prodrome 1–3 days**, then **persistent cough** and **tachypnoea or recession**, and **wheeze or crackles**. Usually **RSV**. Peaks day 3–5; cough resolves in 90% by 3 weeks. |
| 13 | Bronchiolitis management (NICE NG9/WHO). | **Supportive**: oxygen if $SpO_2$ <90–92%, small frequent or **NG feeds**, gentle nasal suction, CPAP if failing. **No** routine antibiotics, steroids, salbutamol, adrenaline or hypertonic saline. (UCG 2023 allows a salbutamol trial, stop if no response.) |
| 14 | Severe bronchiolitis criteria (UCG/NICE). | **Age <3 months**, RR >60–70, poor feeding (<50–75% of normal), **$SpO_2$ <92%**, apnoea, grunting, cyanosis, exhaustion. |

| # | Question | Answer |
|---|---|---|
| 15 | Clinical stages of pertussis. | **Catarrhal** (1–2 wk, most infectious, coryza) → **Paroxysmal** (1–6 wk: paroxysms, whoop, post-tussive vomiting, subconjunctival haemorrhage; infants: **apnoea, cyanosis**, may not whoop) → **Convalescent** (weeks–months). |
| 16 | Antibiotic treatment of pertussis. | **Azithromycin**: <6 months 10 mg/kg once daily × 5 days; ≥6 months 10 mg/kg day 1 then 5 mg/kg days 2–5. Or **erythromycin 12.5 mg/kg 6-hourly** × 7 days (UCG; 14 classically). Avoid erythromycin <1 month (pyloric stenosis). Treat household contacts. |
| 17 | Drugs NOT to give in pertussis (UCG 2023). | **Cough mixtures, sedatives, mucolytics, antihistamines** — 'useless' and potentially harmful. Give oxygen, suction, small frequent feeds and a macrolide. |
| 18 | PCP treatment in an HIV-infected infant (UCG 2023). | **Cotrimoxazole 120 mg/kg/day** (TMP 20 mg/kg/day) in 3–4 doses × **21 days** + **prednisolone 2 mg/kg/day** × 5 days then taper (if hypoxaemic); oxygen; then cotrimoxazole prophylaxis; start ART. |
| 19 | Clinical clues to PCP. | HIV-exposed/infected infant **2–6 months**, not on cotrimoxazole, **severe hypoxaemia** with **few crackles**, fast breathing, afebrile or low fever; CXR diffuse interstitial infiltrates; high LDH. |
| 20 | Empyema: clinical signs and management. | **Persistent fever** on antibiotics, **stony dull percussion**, absent breath sounds, mediastinal shift. CXR/ultrasound, **diagnostic tap**, **chest drain**, cloxacillin + gentamicin (or ceftriaxone) for **3–4 weeks**; exclude TB. |
| 21 | Kerosene ingestion: key management points. | **No induced vomiting, no gastric lavage, no charcoal.** Oxygen if $SpO_2$ <90%, **observe ≥6 h**, CXR if symptomatic; antibiotics only for secondary bacterial pneumonia. |
| 22 | Back blows and thrusts for a choking infant vs child. | **Infant (<1 y)**: 5 back blows (head down along the forearm) + 5 **chest** thrusts. **Child >1 y**: 5 back blows + 5 **abdominal** thrusts. Unconscious → start CPR, look in the mouth before breaths. |
| 23 | Common causes of pneumonia by age (UCG 2023). | **Neonates**: GBS, Klebsiella, E. coli, S. aureus, Chlamydia. **<5 y**: **pneumococcus**, H. influenzae, S. aureus, RSV, influenza, measles. **≥5 y**: pneumococcus, Mycoplasma, viruses. **HIV**: Pneumocystis, TB, CMV. |
| 24 | Why is wheeze important when classifying pneumonia? | Wheeze + fast breathing may be **bronchospasm** (asthma, viral wheeze, bronchiolitis), not pneumonia. Give a rapid-acting bronchodilator and **re-count**; if fast breathing resolves, don't classify as pneumonia. |
| 25 | Reasons a child with pneumonia fails to improve (STEP-HIV-F). | **S**taph, **T**B, **E**mpyema/effusion, **P**CP, **H**eart failure, **I**nadequate dose/adherence, **V**iral/wrong diagnosis, **F**oreign body. Also malaria, anaemia, SAM. |
| 26 | Pneumonia burden (WHO). | **14% of under-five deaths: 740,180 children in 2019**; 22% of deaths at 1–5 y. Pneumonia + diarrhoea ≈ **23% of under-five deaths (1.17 million)** (WHO 2024 guideline). |
| 27 | Vaccines in UNEPI that prevent pneumonia. | **PCV** and **DTP-HepB-Hib** (pertussis, Hib) at **6, 10, 14 weeks**; **measles-rubella at 9 and 18 months**; BCG at birth (severe TB). Plus exclusive breastfeeding and cotrimoxazole for HIV-exposed infants. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A 9-month-old boy (7.5 kg) has had cough and fever for 2 days. He is breastfeeding well and alert. RR 54/min (counted for 60 s while calm), no chest indrawing, no wheeze, $SpO_2$ 96%. Malaria RDT negative. What is the correct classification and treatment?

A No pneumonia: cough or cold; home care without antibiotics
B Severe pneumonia; admit for ampicillin and gentamicin
C Pneumonia; oral amoxicillin 250 mg twice daily for 5 days and review in 2–3 days
D Bronchiolitis; nebulised salbutamol and discharge
E Pneumonia; cotrimoxazole for 5 days

**2.** A 14-month-old girl (8.8 kg) has cough and fever for 3 days. Today she cannot drink. She is lethargic, RR 58/min, lower chest wall indrawing, grunting, $SpO_2$ 87% in air. MUAC 12.8 cm. What is the BEST initial management?

A Oxygen by nasal prongs, then IV ampicillin 440 mg 6-hourly plus gentamicin 66 mg once daily
B Oral amoxicillin 500 mg twice daily at home and review in 2 days
C IV ceftriaxone 2 g once daily as first line
D Nebulised salbutamol and observe for 1 hour
E IV fluids 20 mL/kg bolus, then oral amoxicillin

**3.** A 3-year-old boy (13 kg) with severe pneumonia has received ampicillin and gentamicin for 48 hours. He remains febrile at 39.4 °C with RR 55/min. Chest X-ray shows multiple thin-walled air-filled cysts in the right lung and a small right pleural effusion. He had multiple skin boils last week. What is the MOST appropriate change in antibiotics?

A Continue ampicillin and gentamicin for another 5 days
B Switch to oral amoxicillin
C Add erythromycin for atypical pneumonia
D Start high-dose cotrimoxazole for PCP
E Switch to cloxacillin 650 mg 6-hourly plus gentamicin, continued for at least 3 weeks

**4.** A 4-month-old girl (5.5 kg) has had runny nose for 2 days, then cough and difficulty feeding. She is taking about half her usual breastfeeds. RR 62/min, mild subcostal recession, bilateral fine crackles and expiratory wheeze, T 37.8 °C, $SpO_2$ 89% in air. What is the MOST appropriate management?

A Oral amoxicillin and send home
B Admit, oxygen by nasal prongs, NG feeds or small frequent breastfeeds, gentle nasal suction
C Nebulised salbutamol every 20 minutes and oral prednisolone
D IV ceftriaxone and dexamethasone
E Chest physiotherapy every 4 hours

**5.** At a HC III with no pulse oximeter, which ONE finding in a 10-month-old with pneumonia MOST strongly suggests hypoxaemia and the need to give oxygen and refer?

A Temperature 39 °C
B Respiratory rate 52/min
C Crepitations at the right base
D Head nodding with each breath
E Cough for 4 days

**6.** A 3-month-old HIV-exposed infant (4.6 kg) whose mother did not take ART is admitted with fast breathing (RR 72/min) and $SpO_2$ 78% in air. Temperature 37.6 °C. The chest has only a few scattered crackles. He has never received cotrimoxazole. Which treatment should be added to ampicillin and gentamicin?

- **A** High-dose cotrimoxazole for 21 days plus prednisolone
- **B** Oral amoxicillin 3 days
- **C** Isoniazid preventive therapy
- **D** Fluconazole 6 mg/kg daily
- **E** Nebulised adrenaline

**7.** A 5-week-old girl (3.8 kg), whose DTP doses are not yet due, has had a runny nose for a week and now has bouts of coughing that end with her going blue and stopping breathing for 10–15 seconds, followed by vomiting. Between bouts she looks well and the chest is clear. Her 8-year-old brother has had a cough for 3 weeks. Which antibiotic is MOST appropriate for her? Advanced

- **A** Erythromycin 12.5 mg/kg 6-hourly for 14 days
- **B** Amoxicillin 40 mg/kg twice daily for 5 days
- **C** Azithromycin 10 mg/kg once daily for 5 days
- **D** Ceftriaxone 80 mg/kg once daily for 7 days
- **E** Cotrimoxazole prophylaxis

**8.** A 2-year-old boy (11 kg) drank kerosene from a soda bottle 30 minutes ago. He coughed and vomited once and now has RR 36/min, no indrawing, $SpO_2$ 97%. His father asks you to make him vomit the rest. What is the BEST management?

- **A** Induce vomiting with salt water
- **B** Gastric lavage with normal saline
- **C** Give activated charcoal 1 g/kg
- **D** Start IV ceftriaxone to prevent pneumonia
- **E** Do not induce vomiting; observe for at least 6 hours with $SpO_2$ monitoring and CXR if symptoms develop

**9.** A 4-year-old girl (15 kg) has been treated for severe pneumonia for 4 days. She still has fever 39 °C, RR 48/min, and you find stony dull percussion with absent breath sounds over the left lower chest and the trachea deviated to the right. What is the MOST appropriate next step?

- **A** Add oral salbutamol
- **B** Double the dose of gentamicin
- **C** Discharge on oral amoxicillin
- **D** Chest X-ray/ultrasound and diagnostic pleural tap, then chest drain if pus
- **E** Start anti-TB treatment immediately without further tests

**10.** A 2-year-old boy (11 kg) with cough has RR 46/min and audible expiratory wheeze; he is alert and drinking, $SpO_2$ 95%. What is the MOST appropriate next step before classifying him as pneumonia?

- **A** Start ampicillin and gentamicin
- **B** Give a rapid-acting bronchodilator (salbutamol via spacer) and re-count the respiratory rate after 15–20 minutes
- **C** Do a chest X-ray to confirm consolidation
- **D** Give prednisolone 2 mg/kg and send home
- **E** Classify as severe pneumonia because of wheeze

**11.** A 1-year-old (9 kg) on nasal prong oxygen at 1 L/min for severe pneumonia is now feeding, afebrile and comfortable, RR 38/min. Which is the CORRECT way to decide whether oxygen can be stopped?

A Stop oxygen when the course of antibiotics is finished
B Stop oxygen as soon as the child is afebrile
C Keep oxygen until discharge to be safe
D Increase to 2 L/min for 1 hour, then stop
E Do a trial off oxygen; stop if $SpO_2$ remains ≥90% in room air for 10–15 minutes, then recheck

**12.** A 3-year-old girl (14 kg) with cough has RR 38/min and lower chest wall indrawing. She is alert, drinking well, $SpO_2$ 95%, no danger signs, HIV-negative, no malnutrition, and the family can return. According to the WHO 2024 guideline, what is the recommended treatment?

A Admit for IV benzylpenicillin and gentamicin for 5 days
B Oral amoxicillin 40 mg/kg twice daily for 5 days as an outpatient
C No antibiotic; she has no fast breathing
D Oral amoxicillin for 3 days only
E IM ceftriaxone once and review in a week

**13.** A 6-day-old term baby (3.1 kg) has fast breathing (RR 70/min), severe chest indrawing and poor feeding. Which antibiotic prescription is CORRECT? Advanced

A Oral amoxicillin 40 mg/kg twice daily at home
B Gentamicin 7.5 mg/kg IV once daily alone
C Chloramphenicol 25 mg/kg IV 6-hourly
D Ampicillin 50 mg/kg IV 12-hourly plus gentamicin 5 mg/kg IV once daily
E Cotrimoxazole 4 mg/kg twice daily

**14.** A 7-year-old boy (21 kg) has had cough, fever and night sweats for 5 weeks, with weight loss. His grandmother, who lives with him, is on TB treatment. He completed a course of amoxicillin 2 weeks ago without improvement. RR 30/min, $SpO_2$ 95%. What is the MOST appropriate next step?

A A second course of amoxicillin for 10 days
B Admit for ceftriaxone
C Evaluate for TB: Xpert Ultra on sputum (or gastric aspirate/stool), chest X-ray and HIV test
D Start isoniazid preventive therapy
E Nebulised salbutamol and prednisolone

**15.** A 6-month-old (6.8 kg) with severe pneumonia is on IV fluids because she is too breathless to feed. Her serum sodium is 128 mmol/L and she is not dehydrated. What is the MOST appropriate fluid plan? Advanced

A Restrict fluids to about two-thirds of maintenance and start NG feeds when possible
B Give a 20 mL/kg bolus of normal saline
C Increase fluids to 150 mL/kg/day
D Give 3% saline 10 mL/kg over 10 minutes
E Stop all fluids for 24 hours

## Answers & explanations

**1. C** · Pneumonia; oral amoxicillin 250 mg twice daily for 5 days and review in 2–3 days

C is right: at 2–11 months fast breathing is **≥50/min**; with no danger sign or hypoxaemia this is **pneumonia** → amoxicillin 40 mg/kg (one 250 mg DT) twice daily for 5 days (Uganda is a high-HIV setting) and review in 2–3 days. A is wrong: RR 54 is fast for his age. B is wrong: there are no danger signs, no hypoxaemia and no severe distress. D is wrong: there is no coryza-wheeze-crackles picture and salbutamol isn't standard for bronchiolitis. E is wrong: cotrimoxazole is no longer recommended for pneumonia; amoxicillin is first line.

**2. A** · Oxygen by nasal prongs, then IV ampicillin 440 mg 6-hourly plus gentamicin 66 mg once daily

A is right: a danger sign (lethargy, unable to drink) plus hypoxaemia = **severe pneumonia**: oxygen first, then **ampicillin 50 mg/kg (440 mg) 6-hourly + gentamicin 7.5 mg/kg (66 mg) once daily**. B is wrong: oral outpatient therapy is only for pneumonia without danger signs or hypoxaemia. C is wrong: ceftriaxone is second line, and 2 g is far above 80 mg/kg (≈700 mg) for her weight. D is wrong: there is no wheeze; salbutamol does not treat pneumonia. E is wrong: she is not in shock; boluses risk fluid overload, and oral therapy is inadequate for severe pneumonia.

**3. E** · Switch to cloxacillin 650 mg 6-hourly plus gentamicin, continued for at least 3 weeks

E is right: **pneumatocoeles + effusion + skin boils + failure of first line** = **staphylococcal pneumonia** → **cloxacillin 50 mg/kg (650 mg) 6-hourly + gentamicin** for ≥21 days (UCG 2023), and assess the effusion for empyema. A is wrong: ampicillin does not cover penicillinase-producing Staph, and he is failing. B is wrong: stepping down when worsening is inappropriate. C is wrong: Mycoplasma doesn't cause pneumatocoeles. D is wrong: nothing suggests HIV/PCP, and PCP causes interstitial changes, not cysts with effusion.

**4. B** · Admit, oxygen by nasal prongs, NG feeds or small frequent breastfeeds, gentle nasal suction

B is right: this is **severe bronchiolitis** (coryzal prodrome, crackles and wheeze, feeding about 50% of normal, $SpO_2$ <92%): **supportive care** with oxygen, feeding support (NG if needed) and suction. A is wrong: she is hypoxaemic and not feeding; she needs admission, and antibiotics are not routinely indicated. C is wrong: bronchodilators and steroids don't help bronchiolitis (NICE NG9; UCG allows only a salbutamol trial, stopped if no response). D is wrong: neither antibiotics nor steroids are indicated without evidence of bacterial infection. E is wrong: chest physiotherapy is not recommended in bronchiolitis.

**5. D** · Head nodding with each breath

D is right: **head nodding** (use of accessory neck muscles) is one of the clinical signs WHO uses to predict hypoxaemia when no oximeter is available (with central cyanosis, grunting, nasal flaring, inability to feed, RR ≥70). A is wrong: fever does not indicate hypoxaemia. B is wrong: RR 52 is fast for age but well below the severe tachypnoea threshold (≥70). C is wrong: crackles support pneumonia but do not predict hypoxaemia. E is wrong: cough duration is not a hypoxaemia sign.

**6. A** · High-dose cotrimoxazole for 21 days plus prednisolone

A is right: a 2–6-month-old **HIV-exposed infant** without prophylaxis with **severe hypoxaemia and few chest signs** suggests **PCP** → high-dose cotrimoxazole (UCG 120 mg/kg/day in divided doses for 21 days) with **prednisolone** for hypoxaemia, plus oxygen; test for HIV (DNA PCR) and start ART. B is wrong: oral amoxicillin is for non-severe pneumonia. C is wrong: TPT is for TB prevention after excluding active TB, not acute treatment. D is wrong: fluconazole treats candida/cryptococcus, not PCP. E is wrong: adrenaline is for croup, not pneumonia.

**7. C** · Azithromycin 10 mg/kg once daily for 5 days

C is right: paroxysms with **apnoea, cyanosis and post-tussive vomiting** in a young infant with a coughing sibling = **pertussis**. In infants under 1 month (and preferably under 6 months), **azithromycin 10 mg/kg once daily for 5 days** is preferred. She should be admitted for apnoea monitoring, oxygen and feeding, and household contacts treated. A is wrong: erythromycin in infants <1 month is linked to **infantile hypertrophic pyloric stenosis**; azithromycin is safer. B is wrong: amoxicillin is not active against Bordetella. D is wrong: ceftriaxone is not the treatment for pertussis. E is wrong: cotrimoxazole prophylaxis is for HIV-exposed infants and does not treat pertussis.

**8. E** · Do not induce vomiting; observe for at least 6 hours with $SpO_2$ monitoring and CXR if symptoms develop

E is right: hydrocarbons cause **chemical pneumonitis** when aspirated; management is supportive: **no emesis, no lavage**, observe ≥6 hours, oxygen if $SpO_2$ <90%, CXR if symptomatic. A is wrong: vomiting gives the kerosene a second chance to be aspirated. B is wrong: lavage also risks aspiration. C is wrong: charcoal does not bind hydrocarbons and provokes vomiting. D is wrong: prophylactic antibiotics are not recommended; treat only secondary bacterial pneumonia.

**9. D** · Chest X-ray/ultrasound and diagnostic pleural tap, then chest drain if pus

D is right: persistent fever with **stony dullness, absent breath sounds and mediastinal shift** = large **effusion/empyema**: image, tap (pus? Gram stain, culture, Xpert), and insert a **chest drain**, with antibiotics covering Staph for 3–4 weeks. A is wrong: there is no bronchospasm. B is wrong: doubling gentamicin risks toxicity and does not drain pus. C is wrong: she is getting worse. E is wrong: TB is in the differential of an effusion but must be investigated (tap with Xpert, lymphocytic exudate); acute post-pneumonic empyema is more likely here.

**10. B** · Give a rapid-acting bronchodilator (salbutamol via spacer) and re-count the respiratory rate after 15–20 minutes

B is right: in a wheezing child, fast breathing may be due to **bronchospasm**; WHO/IMCI advise a **bronchodilator trial** and re-assessment before classifying as pneumonia. A is wrong: he has no danger signs or hypoxaemia; injectable antibiotics are not indicated. C is wrong: CXR is not required to diagnose non-severe pneumonia and won't change the immediate step. D is wrong: steroids before a diagnosis (asthma vs viral wheeze vs pneumonia) is premature. E is wrong: wheeze alone is not a severity criterion.

**11. E** · Do a trial off oxygen; stop if $SpO_2$ remains ≥90% in room air for 10–15 minutes, then recheck

E is right: WHO advises a **daily trial off oxygen** when stable; discontinue if $SpO_2$ stays **≥90% in room air** for 10–15 minutes, and recheck after an hour and periodically for the first day. A is wrong: oxygen need is unrelated to antibiotic duration. B is wrong: fever does not determine oxygenation. C is wrong: unnecessary oxygen wastes a scarce resource that other children need. D is wrong: increasing flow before stopping has no rationale.

**12. B** · Oral amoxicillin 40 mg/kg twice daily for 5 days as an outpatient

B is right: WHO (2014, confirmed in the **2024 guideline**) recommends **oral amoxicillin for 5 days as an outpatient** for children 2–59 months with **chest indrawing without danger signs**, rather than injectable antibiotics in hospital. A is wrong: inpatient injectable therapy is for severe pneumonia (danger signs, hypoxaemia). C is wrong: chest indrawing alone classifies as pneumonia. D is wrong: the 3-day option applies only to fast-breathing pneumonia without indrawing, in low-HIV settings. E is wrong: a single ceftriaxone dose is not a recommended regimen and uses a Watch antibiotic.

**13. D** · Ampicillin 50 mg/kg IV 12-hourly plus gentamicin 5 mg/kg IV once daily

D is right: an infant <2 months with pneumonia is treated as **severe**; in the **first week of life** ampicillin is given 12-hourly (50 mg/kg) and **gentamicin 5 mg/kg once daily** (UCG 2023). A is wrong: oral outpatient therapy is not appropriate for a neonate with severe signs. B is wrong: 7.5 mg/kg is the dose after the first week, and gentamicin alone misses GBS and Listeria cover. C is wrong: chloramphenicol is contraindicated in neonates <7 days (grey baby syndrome). E is wrong: cotrimoxazole is not used for neonatal pneumonia and is avoided in the first weeks (kernicterus risk).

**14. C** · Evaluate for TB: Xpert Ultra on sputum (or gastric aspirate/stool), chest X-ray and HIV test

C is right: cough and fever **>2 weeks**, night sweats, **weight loss**, a **household TB contact** and failure of antibiotics = presumptive **TB**; request **Xpert MTB/RIF Ultra** (sputum, induced sputum, gastric aspirate or stool), CXR and an HIV test. A is wrong: repeating antibiotics delays the diagnosis. B is wrong: he is not severely ill and the problem is not bacterial pneumonia. D is wrong: TPT is only given after active TB has been excluded; giving isoniazid alone to a child with TB disease breeds resistance. E is wrong: there is no evidence of asthma.

**15. A** · Restrict fluids to about two-thirds of maintenance and start NG feeds when possible

A is right: hyponatraemia without dehydration in severe pneumonia suggests **SIADH**; restrict to about **two-thirds of maintenance**, use isotonic fluids and move to NG/breast milk early. B is wrong: she is not shocked or dehydrated; a bolus worsens overload. C is wrong: high volumes worsen hyponatraemia and pulmonary oedema. D is wrong: hypertonic saline is reserved for symptomatic severe hyponatraemia (seizures), not a stable infant at 128. E is wrong: stopping all fluids risks hypoglycaemia and dehydration.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Severe pneumonia in an HIV-exposed infant

Namubiru, a 10-month-old girl from Nansana, is brought to Mengo Hospital with 3 days of cough and fever. Today she refused the breast. Her mother is HIV-positive on TLD; Namubiru had a negative DNA PCR at 6 weeks and is still breastfeeding.

**1. What are your first actions in the emergency room?**

> ETAT: check airway/breathing (cyanosis, severe distress), circulation, consciousness; **$SpO_2$**; weigh; RBS; malaria RDT. Give **oxygen** if $SpO_2$ <90% or emergency signs; position on mother's lap; gentle suction.

**2. Findings: 7.2 kg (WHZ –1.5, MUAC 12.4 cm, no oedema), lethargic, RR 64, lower chest wall indrawing, grunting, $SpO_2$ 86%, T 39 °C, bronchial breathing right upper zone, RDT negative, RBS 4.8. Classify.**

> **Severe pneumonia with hypoxaemia** (danger signs: lethargy and unable to breastfeed; $SpO_2$ <90%; grunting).

**3. Write the admission orders.**

> 1. **Oxygen** by nasal prongs 1 L/min, target $SpO_2$ ≥90% (≥94% initially if emergency signs).
> 2. **Ampicillin 360 mg IV 6-hourly** (50 mg/kg) + **gentamicin 54 mg IV once daily** (7.5 mg/kg).
> 3. **Paracetamol 108 mg** (15 mg/kg) 6-hourly prn.
> 4. **NG expressed breast milk** (avoid IV fluids if possible); maintenance about 100 mL/kg/day.
> 5. Monitor RR 2-hourly, $SpO_2$ 3-hourly, temperature 6-hourly, feeding, alertness.

**4. What about her HIV status?**

> She is **HIV-exposed and still breastfeeding**, so a negative 6-week PCR does not exclude infection. Do a repeat test now (rapid antibody at ≥9 months; DNA PCR if reactive). If positive or severe hypoxaemia without response: add **high-dose cotrimoxazole + prednisolone** for PCP, start ART. Ensure cotrimoxazole prophylaxis continues until HIV is excluded after breastfeeding ends.

**5. At 48 hours she is afebrile, feeding by breast, RR 44, $SpO_2$ 93% on 0.5 L/min. Next steps?**

> Improving: continue IV antibiotics to complete ≥3 days then switch to **oral amoxicillin 40 mg/kg (≈300 mg) twice daily** to complete at least 5 days total; **daily trial off oxygen** (stop if $SpO_2$ ≥90% in air for 10–15 min); feeding counselling.

**6. What must happen before discharge?**

> Off oxygen ≥24 h, feeding well; complete antibiotics; **HIV result** and linkage; **catch-up vaccines** (MR1 given? PCV complete?); nutritional follow-up (WHZ –1.5); counselling on danger signs, smoke and breastfeeding; review in 1 week.

**Take-away:** Danger sign plus hypoxaemia = severe pneumonia: oxygen first, then ampicillin + gentamicin by weight, careful feeding, and never leave an HIV-exposed infant's status unresolved.

## Case 2: The child who did not get better

Okello, a 3-year-old boy (13 kg) from Gulu, was admitted 2 days ago with severe pneumonia and is on ampicillin and gentamicin. He had measles 3 weeks ago. The nurse reports he is still febrile and breathing fast.

**1. List the reasons a child may fail first-line treatment.**

> **STEP-HIV-F**: Staph (especially after measles), TB, Empyema/effusion, PCP (HIV), Heart failure, Inadequate dosing/adherence, Viral or wrong diagnosis, Foreign body; plus malaria, severe anaemia, SAM.

**2. Examination: T 39.6 °C, RR 56, $SpO_2$ 89% on 2 L/min, stony dullness and absent breath sounds right base. CXR: right effusion and two left-sided pneumatocoeles. Interpretation?**

> **Staphylococcal pneumonia with empyema** (post-measles, pneumatocoeles, effusion, failure of ampicillin).

**3. What procedures and antibiotics do you order?**

> Ultrasound-guided **diagnostic tap** (pus → Gram stain, culture, Xpert, pH) and **chest drain** with underwater seal. Switch to **cloxacillin 650 mg IV 6-hourly** (50 mg/kg) **+ gentamicin 98 mg once daily** (7.5 mg/kg), for **≥3 weeks** (IV until afebrile, then oral). Increase oxygen; analgesia.

**4. Two days later he suddenly becomes more breathless, $SpO_2$ falls to 80%, and the right chest is hyper-resonant. What has happened?**

> **Pyopneumothorax** (ruptured pneumatocoele or bronchopleural fistula). Check the drain (blocked? displaced?), needle decompression if tension, reposition/insert a new drain, high-flow oxygen, senior surgical review.

**5. What else should you address because of his measles?**

> **Vitamin A** (if not given: 200,000 IU for ≥12 months, on 2 consecutive days), nutritional assessment (measles → malnutrition), HIV test, check eyes, and **catch-up vaccines** (MR2, PCV) after recovery.

> **Take-away:** A child not better at 48 hours is a diagnostic problem: look for Staph (post-measles, pneumatocoeles), empyema, TB and HIV, and change both the antibiotic and the plan.

## Case 3: Apnoea in a young infant

Baby Ruth, 6 weeks old (4.0 kg), is brought to a HC IV because she 'stops breathing and turns blue when she coughs'. She had a runny nose 10 days ago. Her 9-year-old sister has had a cough for 3 weeks that makes her vomit. Ruth received BCG and OPV0 at birth; her 6-week vaccines are due today.

**1. What is the most likely diagnosis and what supports it?**

> **Pertussis**: catarrhal prodrome, then paroxysms with **apnoea and cyanosis** (young infants may not whoop), a sibling with a prolonged cough and post-tussive vomiting, and no DTP yet.

**2. What are your immediate management priorities?**

> **Admit** (age <6 months with apnoea). Oxygen if $SpO_2$ <90% or cyanosis; **gentle suction** after paroxysms; nurse near the nursing station with an apnoea/$SpO_2$ monitor; **small frequent breastfeeds**, NG feeds if needed; keep warm; check glucose.

**3. Which antibiotic and dose?**

> **Azithromycin 10 mg/kg = 40 mg orally once daily for 5 days** (preferred under 1 month and safe at 6 weeks). Erythromycin 12.5 mg/kg 6-hourly is the UCG option but is linked to pyloric stenosis in young infants. Isolate until 5 days of treatment.

**4. What will you NOT give, and why?**

> No **cough mixtures, sedatives, mucolytics or antihistamines** (UCG: useless; sedatives worsen apnoea).

**5. What do you do for the family and the community?**

> **Treat the sister and household contacts** with a macrolide; give Ruth her **6-week vaccines** (pentavalent, PCV, OPV, rotavirus) once stable; check the sister's immunisation card; **notify** the district surveillance focal person (pertussis is notifiable); explain that the cough may last up to 3 months.

**Take-away:** Young infants with pertussis present with apnoea rather than a whoop; admit, support breathing and feeding, give azithromycin, and protect the household with prophylaxis and vaccination.

# References


1. Guideline on management of pneumonia and diarrhoea in children up to 10 years of age. *World Health Organization*, 2024. https://www.who.int/publications/i/item/9789240103412
2. Uganda Clinical Guidelines 2023 (5.2.1 Bronchiolitis; 5.2.8 Pertussis; 5.2.9 Pneumonia; 3.1.10.4 Pneumocystis pneumonia). *Ministry of Health Uganda*, 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
3. Pocket Book of Hospital Care for Children, 2nd edition. *World Health Organization*, 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
4. Pneumonia in children: fact sheet. *World Health Organization*, 2022. https://www.who.int/news-room/fact-sheets/detail/pneumonia
5. Bronchiolitis in children: diagnosis and management (NG9). *National Institute for Health and Care Excellence*. https://www.nice.org.uk/guidance/ng9
6. Bronchiolitis: Clinical Practice Guideline. *The Royal Children's Hospital Melbourne*. https://www.rch.org.au/clinicalguide/guideline_index/Bronchiolitis/
7. Pediatric Pneumonia. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/967822-overview
8. Bronchiolitis. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/961963-overview
9. Pertussis. *Medscape (eMedicine)*. https://emedicine.medscape.com/article/967268-overview


A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.